Lesson Overview: Spark Deeper Concepts

In today's digital landscape, understanding Apache Spark is essential for handling large-scale data. Mastering Spark can accelerate our problem-solving capabilities and enhance our ability to drive decisions in data-intensive environments.

In this lesson, we'll explore the extensive capabilities of Apache Spark for large-scale data processing, with a focus on mastering the Extract, Transform, Load (ETL) process and optimizing workflows.



What You'll Learn:

▼Spark Data Structures

Understand and utilize fundamental Spark data structures in the ETL process.

▼UDFs and Pandas UDFs

Create and apply UDFs and Pandas UDFs for custom data transformations.

▼SparkSQL

Gain proficiency in SparkSQL to manipulate and query datasets effectively.

▼Data Operations Within Spark

Implement advanced data operations such as unions, joins, and windowing within Spark.

▼Optimizing Spark

Optimize the performance of Spark applications, focusing on enhancing ETL and broader data engineering processes.

▼Manage and Transform Data

Manage and transform data effectively by reading data into Spark, implementing projections and filters, modifying data frames, and performing aggregations.

▼Writing Data from Spark to other Systems

Ensure the persistence of data modifications by writing data from Spark to other systems.

▼Connecting Spark to External SQL DBs

Understand the connectivity and data flow between Spark and external SQL databases via JDBC.